

Jimmy Zheng

University of Waterloo • Data Science

 Portfolio Website
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EDUCATIONAL BACKGROUND

University of Waterloo, 2B, Waterloo, Ontario

Sep 2024 - April 2029

- Bachelor of Mathematics, Data Science GPA: 3.7/4.0
- Relevant Coursework: Object-Oriented Programming, Calculus, Algorithm Design & Data Structures, Linear Algebra

TECHNICAL SKILLS

- Languages: Python, JavaScript/TypeScript, Java, Go, C/C++, HTML, CSS, SQL, Dart, Racket, VBA, LabVIEW SDK
- Frameworks/Libraries: React/React Native, Vue.js, Node.js, Express.js, Next.js, NestJS, REST API, Flutter, .NET
- Tools: Postman, Linux CLI, Stripe, GitHub/GitLab, n8n, Render, Cloudflare, Docker, SSH, Cursor, Supabase, Vercel

WORK EXPERIENCE

Full-Stack Developer, Co-op

Jan 2026 - April 2026

Hanov Solutions Inc., Waterloo, Ontario

(Python, Go, Vue.js, Node.js, TypeScript, SSR, pSEO, SQLite, Stripe)

- Engineered and deployed core infrastructure for [eh-trade](#), a momentum-based stock analytics SaaS platform, transforming **Python ML** trading models into a production-ready **full-stack** system serving real-time stock screening.
- Optimized **Vue 3** [charting](#) and [pair analysis](#) interfaces by refactoring **data-heavy frontend** modules, reducing redundant **API** and **RPC** calls, and implementing **lazy loading**, cutting load times by **30%** across desktop and mobile.
- Built a scalable **programmatic SEO** (pSEO) system using **Go**, implementing **server-side rendering** (SSR) for dynamic stock pages, automated **meta/OG tag** generation, **cache-backed data** prewarming, and **sitemap** integration, enabling efficient **indexing** and fast crawl performance across **500+** [earnings pages](#) and hundreds of comparison URLs.
- Developed secure **authentication** and **subscription** infrastructure using **TypeScript**, **Go**, and **Stripe**, enabling recurring billing, gated premium features, and full user lifecycle management from onboarding to subscription control.
- Used **AI-assisted** development workflows (Cursor, Claude) to speed implementation of full-stack features including **charting** improvements, **watchlist** performance, **subscription** flows, and analytics tooling across **Vue.js**, **TypeScript**, **Go**, and **Python**, increasing feature throughput by **40%** without compromising review quality or system reliability.

Software Innovation Developer, Co-op

May 2025 - August 2025

Emerson, NI, Shanghai, China

(C++, Python, LabVIEW SDK, AST, n8n, REST API, RAG, Docker, React)

- Assisted in developing a LabVIEW-to-native-code compiler using **C++ and Python**, enabling **graphical programs** to be translated into machine code. Improved compilation speed by **30%** and reduced runtime errors by **20%**.
- Built automated workflows with **n8n** and **Docker** to generate and publish circuit documentation, reducing manual preparation time by **70%** and ensuring consistent formatting across all projects.
- Designed and built an internal website using **JavaScript** and **ComfyUI** with **React.js** to auto-generate cartoon avatars from employee photos, increasing internal team profile recognizability by **80%**.

Software Developer, Part-time

Feb 2024 - August 2025

VolunTrack.Org, Toronto, Ontario

(Python, React, JavaScript, Jest, Node.js, Express.js, Postman, HTTP)

- Designed and executed structured **API** test cases in **Postman** for **happy-path** flows, **required-field** checks, and **type validation**, verifying correct status codes, response structures, and backend behavior across core endpoints.
- Implemented automated testing using **Python unit tests**, **Jest**, and **visual regression** testing, identifying and resolving **15+** bugs before production release and improving overall system reliability by **50%**.
- Extended test coverage to **authentication** failures, **nonexistent-resource** scenarios, **duplicate submissions**, **rate-limit** behavior, improving backend response accuracy to **95%+** and reducing integration defects across the web application.

PERSONAL PROJECTS

Keeper Note App

Feb 2025 - June 2025

(React, JavaScript, CSS, Async/Await, Postman, HTTP, Unsplash API)

- Built a **production-style** note application with modular components and reusable state management patterns.
- Created a React app with npm scripts + **cross-env** (OpenSSL fix), **ESLint (react-hooks)**, CSS + Google Fonts;
- Composed reusable components (Header, Footer, CreateArea, Note) with **lifted state** and keyed lists.